A Project Report

On

FERTILIZER RECOMMENDATION

For

U23CS506-Artificial Intelligence and Machine Learning Laboratory

Submitted by

ALRIFA I (61782323102007)

DHARSHANA R (61782323102028)

GOPIKA K (61782323102044)

In partial fulfilment for the award of the degree

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

SONA COLLEGE OF TECHNOLOGY (AUTONOMOUS), SALEM- 636005

ANNA UNIVERSITY: CHENNAI 600 025

DECEMBER 2025

SONA COLLEGE OF TECHNOLOGY (AUTONOMOUS)

ANNA UNIVERSITY: CHENNAI - 600 025

BONAFIDE

This is to certify that this project titled "**FERTILIZER RECOMMENDATION**" is Bonafide work of "**ALRIFA I (61782323102007) , DHARSHANA R (61782323102028) , GOPIKA K (61782323102044)"** who carried out the project work for U23CS506 -ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING.

Signature of Lab Incharge	**Head of the Department**
Mrs. S. PADMAVATHY, B.E., M.E.	**Dr. B. SATHYABHAMA, M.TECH.,Ph.D.**
ASSISTANT PROFESSOR,	**PROFESSOR,**
SUPERVISOR	**HEAD OF THE DEPARTMENT**
Department of CSE,	Department of CSE,
Sona College of Technology,	Sona College of Technology,
Salem – 636005.	Salem – 636005.

Submitted for the Project Viva-Voce Examination held on ______________

INTERNAL EXAMINER EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We feel honoured to place warm salutations to the **SONA COLLEGE OF TECHNOLOGY** and the **DEPARTMENT OF CSE,** who has given us the opportunity to obtain the chaired goal of being a professional.

We are deeply indebted and express our profound thanks to our honourable Principal **Dr. S.R.R. SENTHIL KUMAR, M.E., Ph.D.,** Sona College of Technology, Salem, for his valuable support and encouragement.

We express my heartfelt thanks to **Dr. B. SATHIYABHAMA, M.TECH., Ph.D.,** Professor and Head, Department of Computer Science and Engineering, Sona College of Technology, Salem for inspiration and valuable suggestions for carrying out this endeavour.

We are extremely thankful to **Mrs. S. PADMAVATHY, B.E., M.E.,** Assistant Professor, Department of Computer Science and Engineering, for her enthusiastic and persistent guidance rendered throughout the completion of the work.

We are also grateful to all **Teaching and Non-Teaching faculty members** of the Computer Science and Engineering Department and friends for their help and co-operation in the successful completion of this project work.

TABLE OF CONTENT

ABSTRACT

This project addresses the critical need for precision nutrient management in modern agriculture by developing an intelligent system capable of predicting the optimal fertilizer requirement. Deciding the correct fertilizer (classified by name, e.g., Urea, DAP, MOP) is a complex challenge, dependent on the dynamic interplay of soil chemistry (N, P, K), environmental conditions (Temperature, Moisture, Humidity), and contextual factors (Soil Type, Crop Type). Traditional methods are either prone to inaccuracy or burdened by the significant time lag of lab testing.

To overcome these limitations, this study utilizes a comprehensive, multi-variable dataset to build a robust machine learning classification model. The data underwent rigorous preprocessing, including class balancing (Oversampling) and feature engineering—specifically creating the Temperatrue Moisture interaction term to enhance environmental context.

The Hist Gradient Boosting Classifier was selected and trained within a Scikit-learn Pipeline, which manages all necessary transformations (scaling and encoding) consistently. This ensemble model demonstrated superior performance in accurately classifying the required fertilizer type. Crucially, the final pipeline was serialized using the joblib library. This step decouples the resource-intensive training process from the user interface, allowing the Streamlit application to load the entire model instantly from disk, resulting in sub-second prediction latency.

This system delivers real-time, data-driven fertilizer recommendations using a high-accuracy Hist Gradient Boosting Classifier in a Streamlit app, ensuring sub-second prediction latency. Targeting Tamil Nadu's agriculture, it minimizes wasteful use of subsidized fertilizers like urea, easing the state's financial burden. By improving nutrient efficiency, it boosts small farmers' profitability and reduces runoff, supporting long-term environmental sustainability.